

PACKAGE INSERT

SUNVASC 5 TABLET

Each tablet contains:
Amlodipine besylate
(equivalent to Amlodipine 5mg) 6.934mg

Product Description: Round, flat, white, plain tablet with single score on one side only.

Pharmacodynamics:
Amlodipine is a dihydropyridine calcium channel-blocker and inhibits the transmembrane influx of calcium ions into the cardiac and vascular smooth muscle.
The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on the vascular smooth muscle. The precise mechanism by which amlodipine relieves angina has not been fully determined, but amlodipine reduces the total ischemic burden by the following 2 actions:
1) Amlodipine dilates the peripheral arterioles and thus, reduces the total peripheral resistance (afterload) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.
2) The mechanism of action of amlodipine probably involves dilation of the main coronary arteries and coronary arterioles, both in normal and ischemic regions. This dilatation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina) and blunt smoking-induced coronary vasoconstriction.

Pharmacokinetics:
After oral administration of therapeutic doses, amlodipine is well absorbed with peak blood levels between 6-12 hours post-dose. Absolute bioavailability has been estimated to be between 64 and 80%. The volume of distribution was approximately 21 L/kg. Absorption of amlodipine is unaffected by consumption of food. The terminal plasma elimination half-life is about 35-50 hours and is consistent with once-daily dosing. Steady-state plasma levels are reached after 7-8 days of consecutive dosing. Amlodipine is extensively metabolized by the liver to inactive metabolites with 10% of the parent compound and 60% of metabolites excreted in the urine.
In vitro studies have shown that approximately 97.5% of circulating amlodipine is bound to plasma proteins.

Indication:
First-line treatment of hypertension and as the sole agent to control blood pressure in the majority of patients. Amlodipine has also been used successfully in combination with a thiazide diuretic, β -adrenoceptor-blocking agent or an angiotensin-converting enzyme inhibitor.
Amlodipine is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Amlodipine may be used alone, as monotherapy or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of β -blockers.

Recommended Dosage:
For both hypertension and angina, the usual initial dose is 5mg once daily which may be increased to a maximum dose of 10 mg depending on the individual patient's response.
No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, β -blockers and angiotensin-converting enzyme inhibitors.
Elderly, children, patients with impaired hepatic function and renal failure: Refer "Warnings and Precautions".

Route of Administration: Oral

Contraindications:
Patients with known sensitivity to dihydropyridines.

Warnings and Precautions:
Use in children:
No clinical experience available on use of amlodipine in children. Not recommended.

Use in elderly:
The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated. Therefore, normal dosage regimens are recommended.

Renal Failure:
Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment, therefore the normal dosage is recommended. Amlodipine is not dialysable.

Impaired Hepatic Function:
As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

Interactions with Other Medicaments:
Amlodipine has been safely administered with thiazide diuretics, β -blockers, angiotensin-converting enzyme inhibitors, long-acting nitrates, sublingual nitroglycerin, NSAIDs, antibiotics and oral hypoglycemic drugs. Co-administration of cimetidine did not alter the pharmacokinetics of amlodipine.
In vitro data from studies with human plasma indicate that amlodipine has no effect on protein-binding of digoxin, phenytoin, warfarin or indomethacin.

Pregnancy And Lactation:
Safety of amlodipine in human pregnancy or lactation has not been established. Hence, use in pregnancy is only recommended when there is no safer alternative and when the disease itself carries greater risk for the mother and child.

Side Effects:
Amlodipine is well tolerated. In placebo-controlled clinical trials involving patients with hypertension or angina, the most commonly observed side effects were headache, edema, fatigue, nausea, abdominal pain, flushing, palpitations, somnolence and dizziness. Less commonly observed side effects in marketing experience include dry mouth, vomiting, increased urinary frequency, alopecia, purpura, skin discoloration, increased sweating, arthralgia, myalgia, tinnitus, visual disturbances, tremor, syncope, paraesthesia, pruritus, rash, dyspnea, rhinitis, asthenia, muscle cramps, dyspepsia, gingival hyperplasia and rarely, erythema multiforme. As with other calcium-channel blockers, the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease, eg myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.
No pattern of clinically significant laboratory test abnormalities related to amlodipine has been observed.

Symptoms and Treatment of Overdose:
Available data suggest that gross overdose could result in excessive peripheral vasodilation and possibly reflex tachycardia. Marked and probably prolonged systemic hypotension up to and including shock with fatal outcome have been reported.
Gastric lavage may be worthwhile in some cases. Clinically significant hypotension due to amlodipine overdose calls for active cardiovascular support including monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. IV calcium gluconate may be beneficial in reversing the effects of calcium-channel blockade. Since amlodipine is highly protein-bound, dialysis is not likely to be of benefit.

Storage Conditions: Store below 30°C.

Pack Size: A bottle of 30 and 60 tablets.
Blister pack: 3 x 10 , 10 x 10 and 50 x 10 tablets per strip.
Pack Size (export only): A bottle of 100 tablets.

Shelf-life: 3 years.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.

Manufacturer and Product Registration Holder:
SUNWARD PHARMACEUTICAL SDN. BHD.
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